Inner product

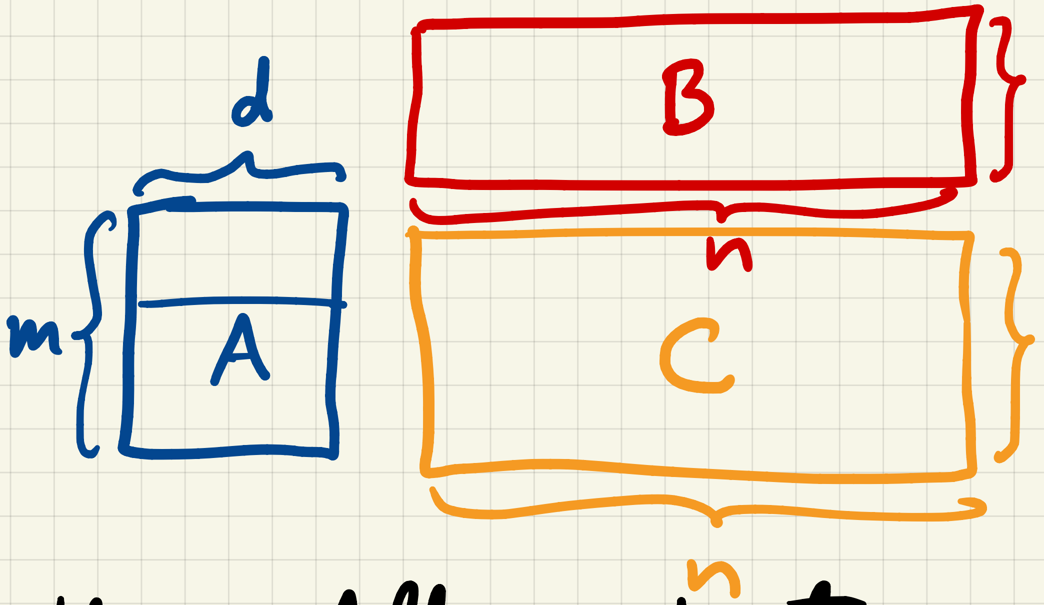
$$\begin{array}{cccccc} & [1, & -3, & 2, & \dots & 4] \\ \times & [2, & 0, & -1, & \dots & 2] \\ \hline & 1\times2 & -3\times0 & 2\times-1 & \dots & 4\times2 \end{array}$$

$$2 + 0 + -2 + \dots + 8 = \boxed{}$$

Use: similarities & compatibilities

Apps: customer & prod
customer & customers

Matrix mult.

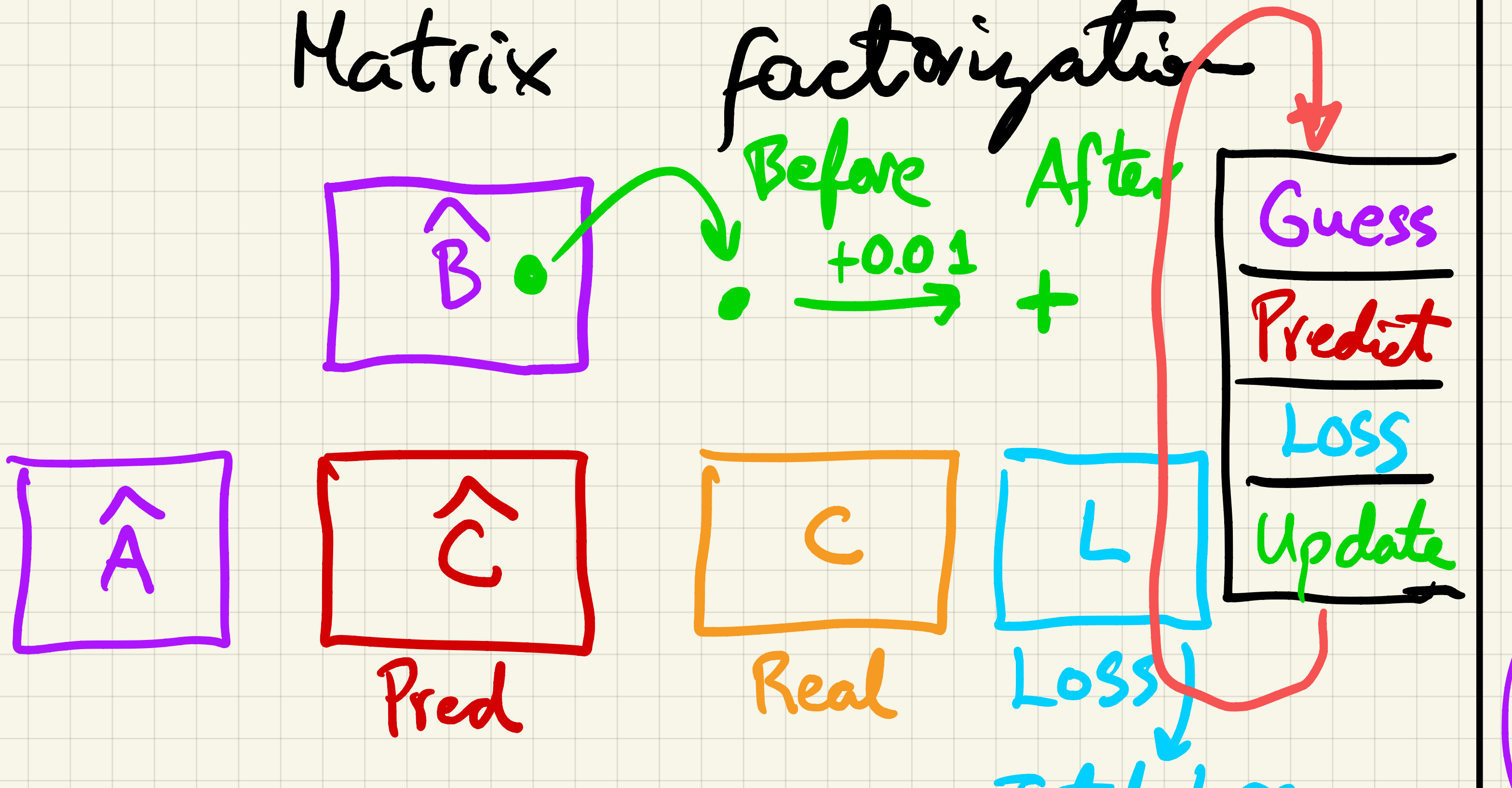


C_{ij} = row i of A × col j of B

Use: All similarities

Apps: All customer & all prods

Matrix factorization

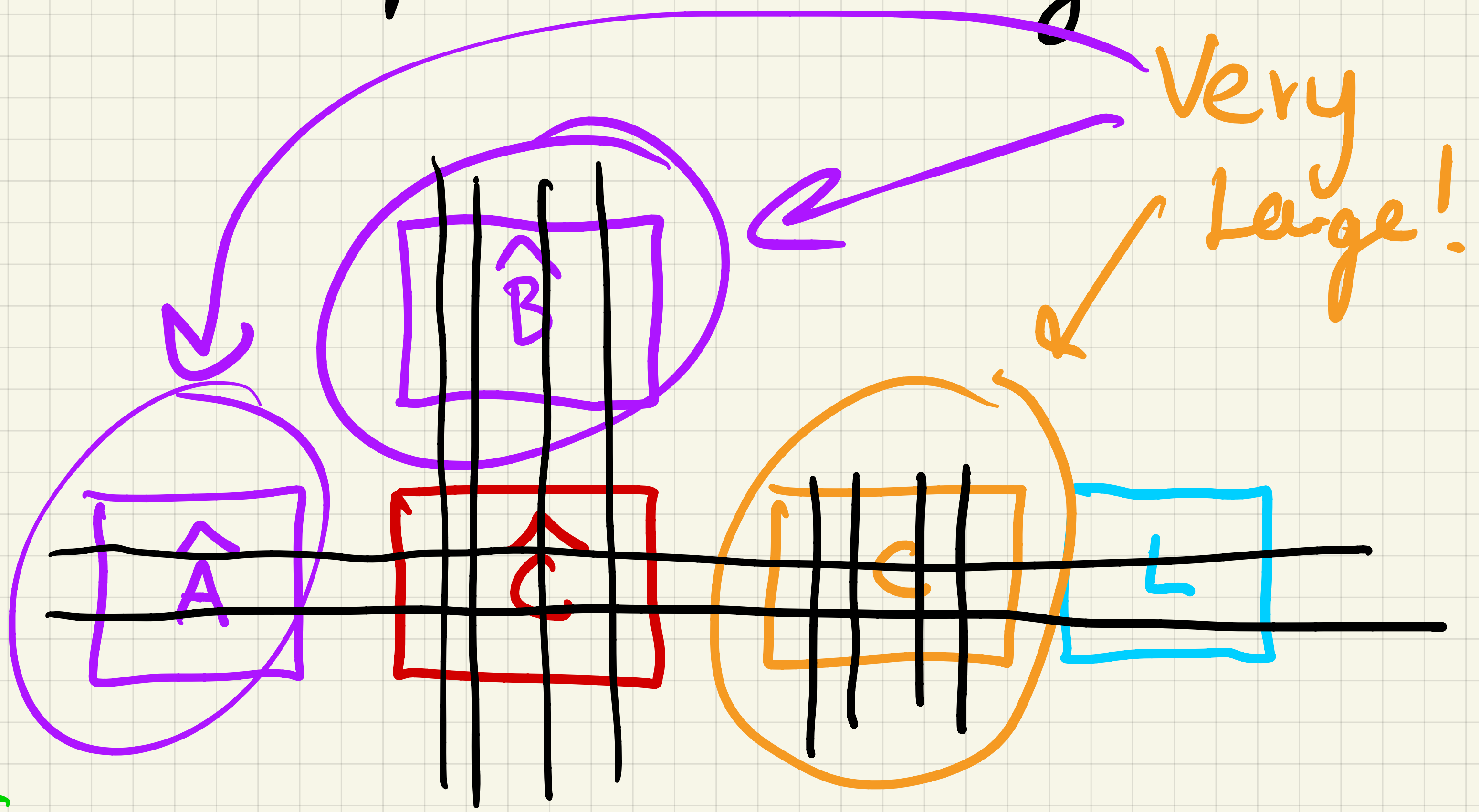


B̂
Before
After
+0.01
Guess
Predict
Loss
Update
Â
Ĉ
C
L
Pred
Real
Loss
Total loss
New loss

Use: Estimate embeddings of customers & products

Apps:

Large-scale computing



Very Large!
B̂
Â
Ĉ
C
L

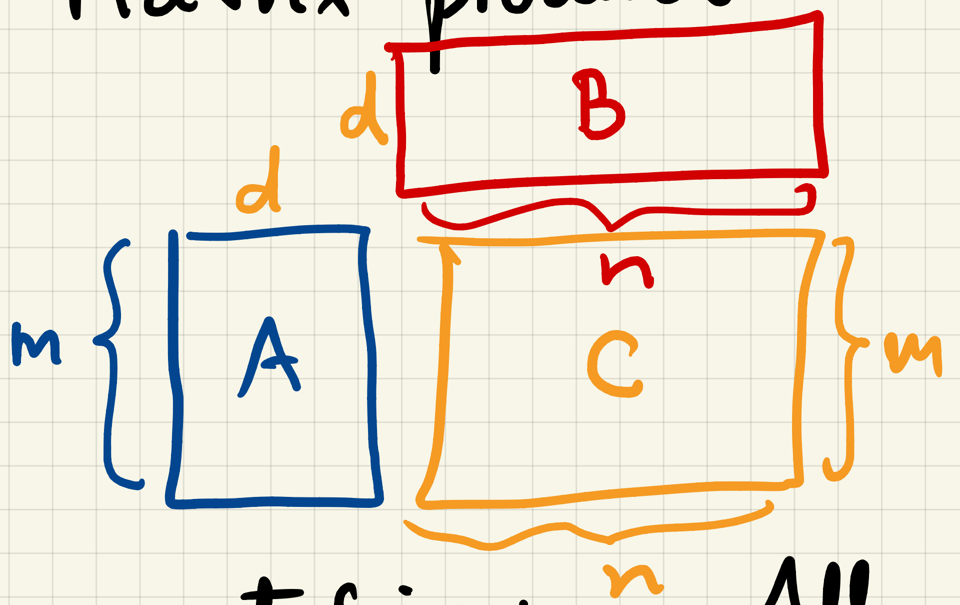
Inner product

$$\begin{array}{r} (x_1, x_2, x_3, \ldots, x_d) \\ \times \quad (y_1, y_2, y_3, \ldots, y_d) \\ \hline x_1y_1 + x_2y_2 + x_3y_3 + x_dy_d \end{array}$$

quantifies: similarity, copatibility

applications: customers & prods
customers & customers

Matrix product

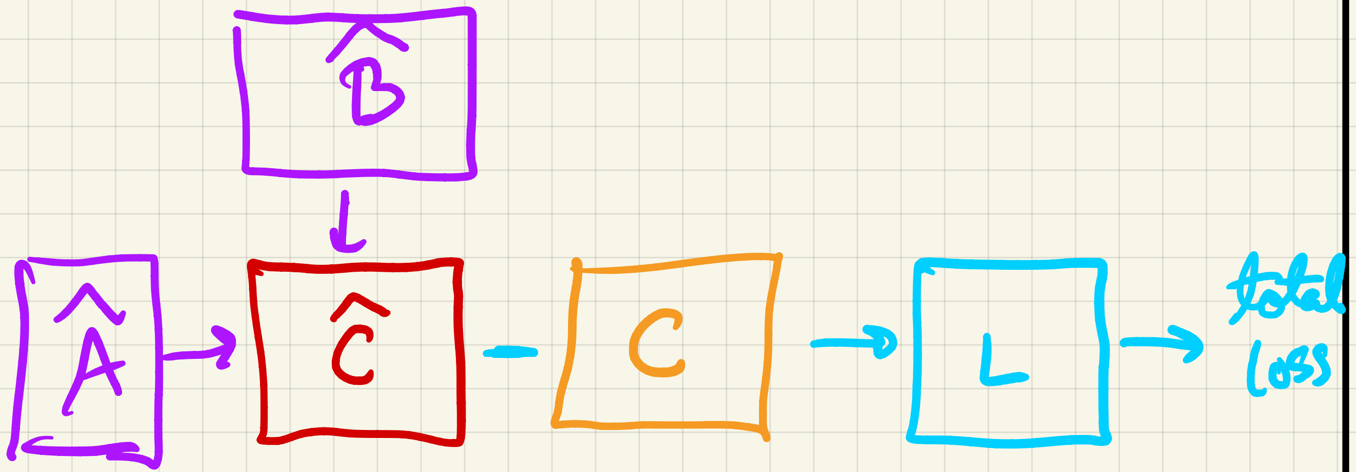


C_{ij} = row i of A × col j of B

quantifies: All similarities

applications: All customers & All prods

Matrix factorization



use: discover embeddings of customers & prods

applications: search, recommendation, pred.

Large-scale cop